MSBA 265 – Foundational Module 1

Interactive EDA, Data Quality Verification, Data Dictionary & Outlier Pipeline

Module 1 Report

Dataset	French Motor Third Party Liability Claims (freMTPL2freq)
Source	OpenML ID 41214
Records analyzed	678,013
Student	Lucky (NgocMaiNhi Nguyen)
Term	Fall 2026

1. Executive Summary

The French Motor Third Party Liability Claims dataset contains 678,013 policy records and 12 variables. The notebook performs structural checks, boundary audits, business data-dictionary construction, skewness diagnostics, Pearson correlation analysis, distribution visualization, and an outlier audit. The production pipeline applies Tukey's 1.5 × IQR rule to Population Density and produces a cleaned dataset with 600,447 records.

#	Column	DataType	Count
1	IDpol	float64	non-null
2	ClaimNB	int64	non-null
3	Expossure	float64	non-null
4	Area	str	non-null
5	VehPower	int64	non-null
6	VehAge	int64	non-null
7	DrivAge	int64	non-null
8	BonusMalus	int64	non-null
9	VehBrand	str	non-null
10	VehGas	str	non-null
11	Density	int64	non-null
12	Region	str	non-null

Dataset Overview: 678,013 rows × 12 columns, no missing values across any feature.

Code script:

1. Loading Libraries for Data Manipulation and Visualization

```
import os

import pandas as pd

import numpy as np

import matplotlib.pyplot as plt

import seaborn as sns
```

```python
# Set visual styling for clean presentation graphics
plt.style.use('seaborn-v0_8-whitegrid')
plt.rcParams['font.sans-serif'] = ['DejaVu Sans']
plt.rcParams['font.size'] = 10
pd.set_option('display.max_columns', 20)

# Load dataset
data_path = os.path.join("..", "data", "raw_business_data.csv")

if not os.path.exists(data_path):
    data_path = os.path.join("data", "raw_business_data.csv")

df = pd.read_csv(data_path)

print(f"[+] Loaded Dataset Shape: {df.shape[0]:,} rows x {df.shape[1]} columns")

# Render interactive HTML table head
df.head(10)
```

2. STRUCTURAL DATA TYPES & NULLS

```python
df.info()
```

3. SUMMARY STATISTICS (NUMERICAL)

```python
df.describe().T
```

2. Part 2 – Raw Boundary Audits & Business Dictionary

2.1 Structural Data and Boundary Verification

Feature	Minimum	Maximum	Audit conclusion
DrivAge	18	100	No negative-age corruption detected
Exposure	0.002732	2.01	No negative exposure detected
BonusMalus	50	230	No negative index values detected

Negative-value checks in the notebook returned zero for DrivAge, Exposure, and BonusMalus. These checks support the conclusion that the raw dataset contains no negative-age or negative-exposure corruption in the audited fields.

2.2 . Univariate Skewness Diagnostics_Density/Area Defense

Density is a continuous measure of population concentration, while Area represents geographic categories. Because both variables describe geographic information, including both in the **same linear regression** can introduce **redundant spatial information and unnecessary model complexity**. For the current modeling approach, Density is retained as the continuous geographic predictor and Area is removed.

This decision should ultimately be guided by business significance and modeling objectives.

Feature	Skewness Coefficient	Diagnosis	Action Required
ClaimNb	5.60	Extremely Right-Skewed	Log / Box-Cox Transform
Density	4.65	Extremely Right-Skewed	Log / Box-Cox Transform
BonusMalus	1.73	Moderately Skewed	Log / Box-Cox Transform
VehPower	1.17	Moderately Skewed	Log / Box-Cox Transform
VehAge	1.15	Moderately Skewed	Log / Box-Cox Transform

DrivAge	0.44	Unskewed / Symmetric	Standard Scaler Sufficient
IDpol	0.24	Unskewed / Symmetric	Standard Scaler Sufficient
Exposure	0.09	Unskewed / Symmetric	Standard Scaler Sufficient

Why keep Density instead of Area? Density is retained because it provides a continuous numerical representation of population concentration, while Area provides a categorical geographic representation. Using both can duplicate spatial information. The final choice should consider business meaning and the intended model

$$\log(1 + \text{Density})$$

Comparison

Density	**Area**
Density ↓ Extremely right-skewed ↓ log(1 + Density) ↓ **Predictor X3**	Area ↓ Spatial redundancy ↓ **Drop**

Keep in mind: High correlation → detect redundancy/multicollinearity → then consider business significance and modeling objectives to decide which variables to retain.

2.3 Exposure Offset Defense

Exposure represents the length of time that an insurance policy is observed during the policy period. Because policies can have different exposure durations, ClaimNb should be interpreted as a claim rate rather than only a raw claim count. Therefore, log(Exposure) can be used as an offset in a Poisson model so that policies with different exposure periods are compared fairly. Density is a continuous numerical variable, while Area is a categorical geographic variable. Including both MSBA 265 – Foundational Module 1 Page 3 may introduce redundant geographic information and increase model complexity. Therefore, Density is retained and Area is removed for the current modeling approach.

2.4 Business Data Dictionary

The notebook defines business meanings for all 12 fields, including policy ID, claim count, exposure, geographic area, vehicle characteristics, driver age, risk score, population density, and administrative region. The executed dictionary export should also be checked before submission because the current notebook output shows only the Region row in the displayed/exported DataFrame, indicating an indentation issue in the dictionary-building loop.

3. Part 3 – Correlation Visual & ClaimNb Decision

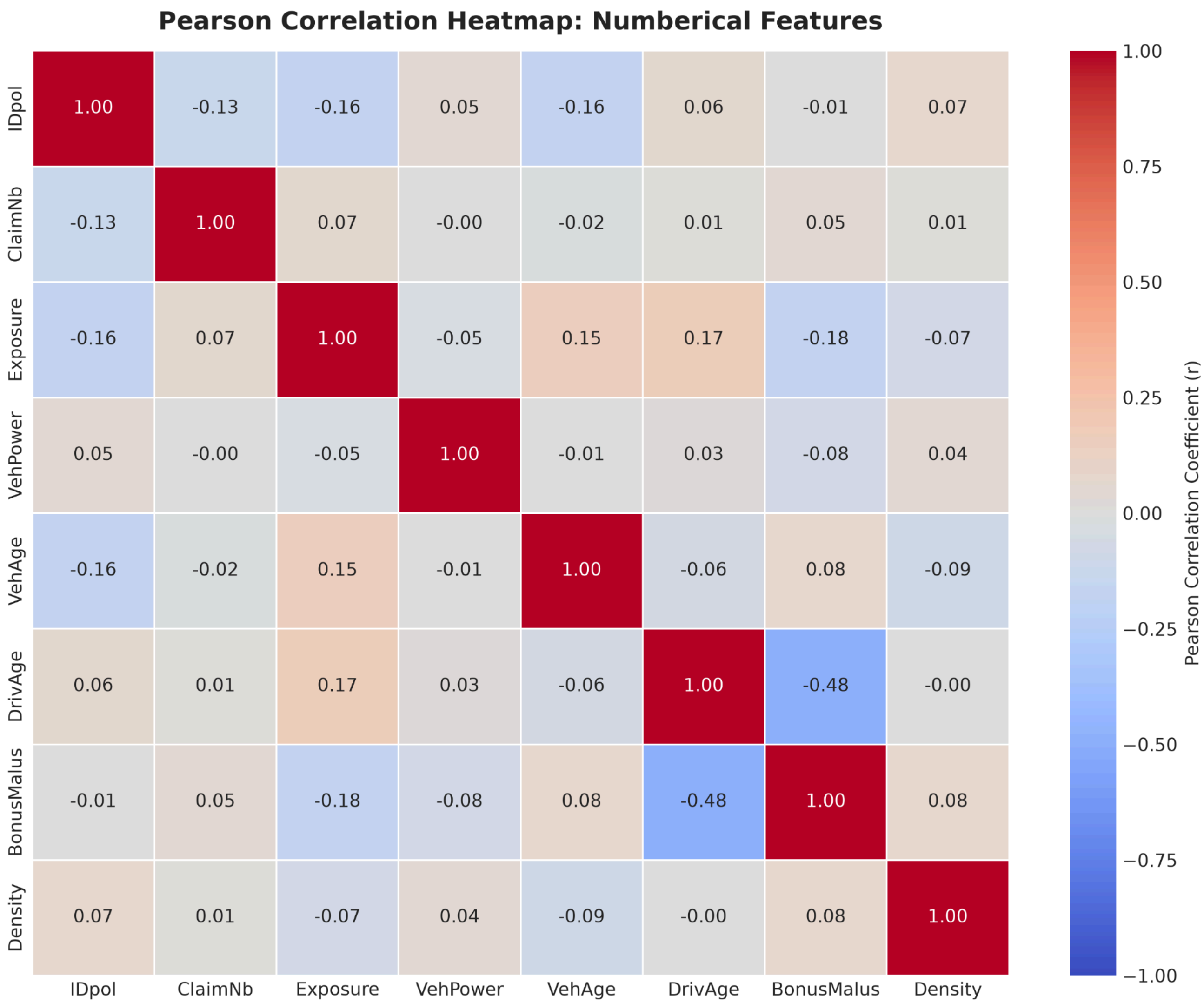

Figure 1. Pearson correlation heatmap generated from the submitted notebook.

What is Pearson correlation purpose?

Pearson r ∈ [-1, +1]

r ≈ +1→ strong positive linear relationship

r ≈ 0 → little/no linear relationship

r ≈ -1→ strong negative linear relationship

Pearson correlation support in providing the prediction which could have risk in serve as a threshold to signal severe multicollinearity.

No severe collinear feature pairs (|r| ≥ 0.85) were detected across the numerical features. The strongest relationship observed was between DrivAge and BonusMalus (r = -0.48), reflecting that older drivers tend to carry lower BonusMalus (risk) scores.

3.1 Heatmap interpretation

• **Relationship Shape:** The numerical features show mostly weak linear relationships, with no severe multicollinearity detected at the $|r| \geq 0.85$ threshold.

• **Key Metrics/Relationships:** The strongest observed relationship is between DrivAge and BonusMalus ($r = -0.48$), indicating a moderate negative relationship in which older drivers tend to have lower BonusMalus scores.

• **Recommended Analytics Action:** No numerical feature pair requires removal based on the $|r| \geq 0.85$ rule. Correlation should still be considered together with business meaning and modeling objectives when selecting predictors.

3.2 ClaimNb Executive Defense

Question: Why ClaimNb also got the "extreme skewness coefficient" but there are no logarithmic transformations conducted?

Although the automated univariate diagnostics flag ClaimNb with an **extreme skewness coefficient** of **approximately 5.60** and suggest a logarithmic transformation, the rule is intentionally overridden.

More than **95% of policyholders record zero claims**, so a natural-log transformation is mathematically impossible for those observations because $\log(0) = -\infty$. ClaimNb is also a discrete count of insurance events rather than a continuous positive-volume measure.

=> ClaimNb should remain in raw count units and can be modeled as a Poisson rate target with log(Exposure) used as an offset.

Code script:

- Report

```
business_definitions = {
    " IDpol ": (" Policy ID", "Key ID", "Unique policy identification key assigned to contract."),
    " ClaimNb ": (" Claim Count ", " Count ", " Total insurance claims filed during policy coverage period ."),
    " Exposure ": (" Risk Exposure ", " Years (0.0 -1.0) ", " Fraction of year policy was active (e.g., 0.50 = 6 months coverage )."),
    " Area ": (" Area Risk Code ", " Ordinal Category ", " Geographic risk code of policyholder address (A = rural , F = urban )."),
    " VehPower ": (" Engine Power ", "Tax HP", " Fiscal power rating category of vehicle engine ."),
```

```
    " VehAge ": (" Vehicle Age", " Years ", "Age of the insured vehicle in years ."),
    " DrivAge ": (" Driver Age", " Years ", "Age of the primary insured driver in years ."),
    " BonusMalus ": (" Claim History Index ", " Index Score (50 -350) ", " Regulatory driver risk
score (< 100: safe discount ; > 100: claim surcharge )."),
    " VehBrand ": (" Brand Class ", " Category Code ", " Anonymized vehicle manufacturer / brand
classification code ."),
    " VehGas ": (" Fuel Type ", " Binary Class ", " Engine fuel type used ( Diesel vs. Regular /
Gasoline )."),
    " Density ": (" Population Density ", " People / km2", " Population density of policyholder ’s
municipality ."),
    " Region ": (" Administrative Region ", " Region Code ", " French administrative region code
corresponding to residency location .")
}
dict_list = []
for col in df.columns :
    label , unit , definition = business_definitions .get (col , (col , "N/A", "No business definition
provided ."))
dict_list . append ({
    " Feature Name ": col ,
    " Business Label ": label ,
    " Data Type ": str(df[col ]. dtype ),
    " Unit ": unit ,
    " Null Count ": df[col ]. isnull ().sum () ,
    " Unique Values ": df[col ]. nunique () ,
    " Business Definition ": definition
})
dict_df = pd. DataFrame ( dict_list )
display ( dict_df )
```

```
reports_dir = os. path . join ("..", "reports")

os. makedirs ( reports_dir , exist_ok = True )

dict_path = os. path . join ( reports_dir , "data_dictionary.csv")

dict_df.to_csv(dict_path, index=False)

print(f"n[+] Sucessfully exported Business Data Dictionary artifact to: {dict_path}")
```

- Skrewness

```
print (" ============================================================ ")

print (" UNIVARIATE SKEWNESS DIAGNOSTICS ")

print (" ============================================================ ")

numeric_df = df. select_dtypes ( include =[ np. number ])

skewness_series = numeric_df . skew (). sort_values ( ascending = False )

skew_summary = pd. DataFrame ({

" Feature ": skewness_series .index ,

" Skewness Coefficient ": skewness_series .values ,

" Diagnosis ": [" Extremely Right - Skewed " if s > 3.0 else " Moderately Skewed " if

abs(s) > 1.0 else " Unskewed / Symmetric " for s in skewness_series . values ],

" Action Required ": ["Log / Box -Cox Transform " if abs(s) > 1.0 else " Standard Scaler Sufficient
" for s in skewness_series . values ]})

display ( skew_summary )
```

- Pearson Multicollinearity audit

```
print (" ============================================================ ")

print (" PEARSON MULTICOLLINEARITY AUDIT ")

print (" ============================================================ ")



corr_matrix = numeric_df . corr (). abs ()
```

```
upper_tri = corr_matrix . where (np. triu (np. ones ( corr_matrix . shape ), k=1). astype ( bool ))


high_corr_pairs = []
for col in upper_tri . columns :high_corr_features = upper_tri . index [ upper_tri [col] >= 0.85].
tolist ()
for feat in high_corr_features :
    val = upper_tri .loc[feat , col]
    high_corr_pairs . append ({
        " Feature 1": feat ,
        " Feature 2": col ,
        " Pearson Correlation (|r|)": round (val , 4) ,
        " Status ": " ACTION REQUIRED : Severe Multicollinearity "
        })


if high_corr_pairs :
    display (pd. DataFrame ( high_corr_pairs ))
else :
    print ("[+] No severe collinear feature pairs (|r| >= 0.85) detected across numerical metrics .")
```

- Pearson correlation Heatmap

```
# Cell 6b: Render Annotated Pearson Correlation Heatmap
plt . figure ( figsize =(10 , 8))
# Compute Pearson correlation matrix across numerical variables
corr = numeric_df . corr ()
# Generate a diverging color map (Cool - Warm : Blue = Negative , Red = Positive )
sns.heatmap(
    corr,
```

```
    annot=True,
    fmt=".2f",
    cmap="coolwarm",
    vmin=-1.0,
    vmax=1.0,
    linewidths=0.5,
    cbar_kws={"label": "Pearson Correlation Coefficient (r)"}
)
plt . title("Pearson Correlation Heatmap: Numberical Features", fontsize=14, fontweight='bold',
pad=12)
plt . tight_layout ()

heatmap_path = os. path . join ("..", "reports", "figures", "correlation_heatmap.png")
plt.savefig(heatmap_path, dpi=300)
plt.show()
print(f"[+] Saved correlation heatmap graphics artifact to: {heatmap_path}")
```

4. Part 4 – Distribution Audits & Written Plot Statements

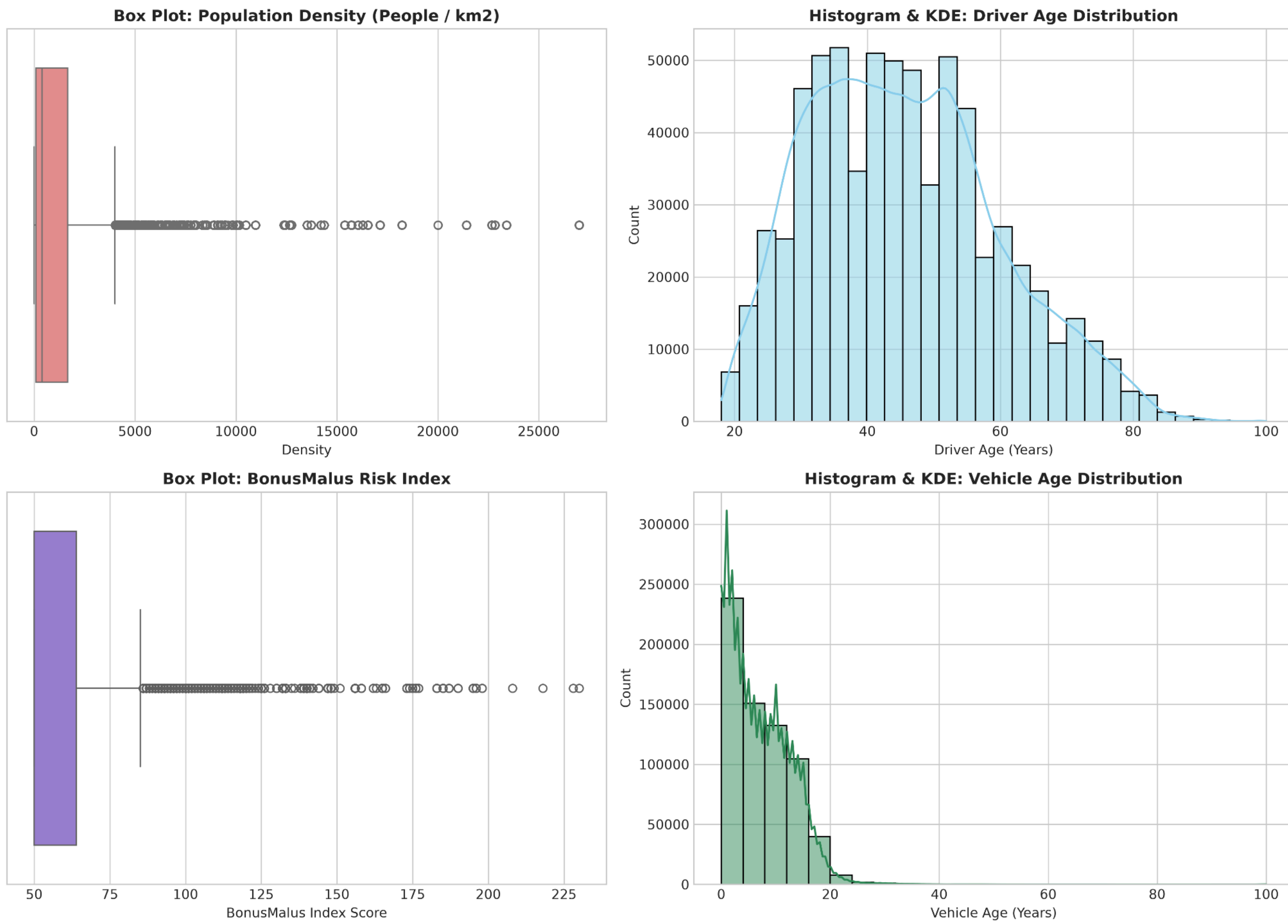

Figure 2. Side-by-side box plot and histogram/KDE distribution audit.

Density – Box Plot

- **Distribution Shape:** Extremely right-skewed; skewness is approximately 4.65.
- **Key Metrics/Whiskers:** The upper Tukey fence is approximately 4,007 people/km²; observations above the fence include high-density metropolitan locations and are not automatically invalid.
- **Recommended Analytics Action:** Use log(1 + Density) before linear regression to reduce the influence of the long right tail; use the Tukey rule for the production cleaning step when an outlier-filtered dataset is required.

DrivAge – Histogram & KDE

- **Distribution Shape:** Approximately symmetric and bell-shaped, with skewness about 0.44.
- **Key Metrics / Whiskers:** Median driver age is 44, with Q1 = 34 and Q3 = 55.
- **Recommended Analytics Action:** No outlier trimming is required; standard Z-score scaling is sufficient if scaling is needed for a linear model.

BonusMalus – Box Plot

- **Distribution Shape:** Moderately right-skewed, with skewness about 1.73.
- **Key Metrics / Whiskers:** The central distribution is concentrated below 100, with Q1 = 50 and Q3 = 68; high values represent drivers with elevated risk scores.
- **Recommended Analytics Action:** Keep the variable continuous and scaled; additionally create an Is Surcharge indicator for BonusMalus > 100 when a business interpretation is useful.

VehAge – Histogram & KDE

- **Distribution Shape:** Moderately right-skewed, with skewness about 1.15.
- **Key Metrics / Whiskers:** The distribution peaks among newer vehicles; Q1 = 2 years and Q3 = 12 years.
- **Recommended Analytics Action:** Bin vehicle age into 0–5, 6–10, and 11+ year groups, or apply standard min-max scaling.

Code Script:

```
fig, axes = plt.subplots(2, 2, figsize=(14, 10))

sns.boxplot(x=df['Density'], ax=axes[0,0], color='lightcoral')

axes[0,0].set_title('Box Plot: Population Density (People / km2)', fontsize=12, fontweight='bold')

axes[0,0].set_xlabel('Density')


sns.histplot(df['DrivAge'], kde=True, ax=axes[0,1], color='skyblue', bins=30)

axes[0,1].set_title('Histogram & KDE: Driver Age Distribution', fontsize=12, fontweight='bold')

axes[0,1].set_xlabel('Driver Age (Years)')


sns.boxplot(x=df['BonusMalus'], ax=axes[1,0], color='mediumpurple')

axes[1,0].set_title('Box Plot: BonusMalus Risk Index', fontsize=12, fontweight='bold')

axes[1,0].set_xlabel('BonusMalus Index Score')


sns.histplot(df['VehAge'], kde=True, ax=axes[1,1], color='seagreen', bins=25)

axes[1,1].set_title('Histogram & KDE: Vehicle Age Distribution', fontsize=12, fontweight='bold')

axes[1,1].set_xlabel('Vehicle Age (Years)')
```

```
plt.tight_layout()

fig_dir = os.path.join("..", "reports", "figures")
os.makedirs(fig_dir, exist_ok=True)         # <-- creates the folder(s) if missing
fig_path = os.path.join(fig_dir, "feature_distributions.png")
plt.savefig(fig_path, dpi=300)
plt.show()
print(f"[+] Saved distribution graphics artifact to: {fig_path}")
```

5. Part 5 – Production Pipeline & Final PDF Export

5.1 Production Outlier Pipeline

The submitted production script src/clean_outliers.py reads the raw business data, calculates Q1 and Q3 for Density, computes IQR = Q3 − Q1, and applies Tukey's 1.5 × IQR lower and upper bounds. Records outside the valid Density interval are removed, and the resulting data are written to data/cleaned_business_data.csv.

Pipeline metric	**Result**
Target feature	Density
Initial dataset	678,013 records
Tukey IQR lower bound	-2,257.00
Tukey IQR upper bound	4,007.00
Outlier records removed	77,566 (11.44%)
Final cleaned dataset	600,447 records

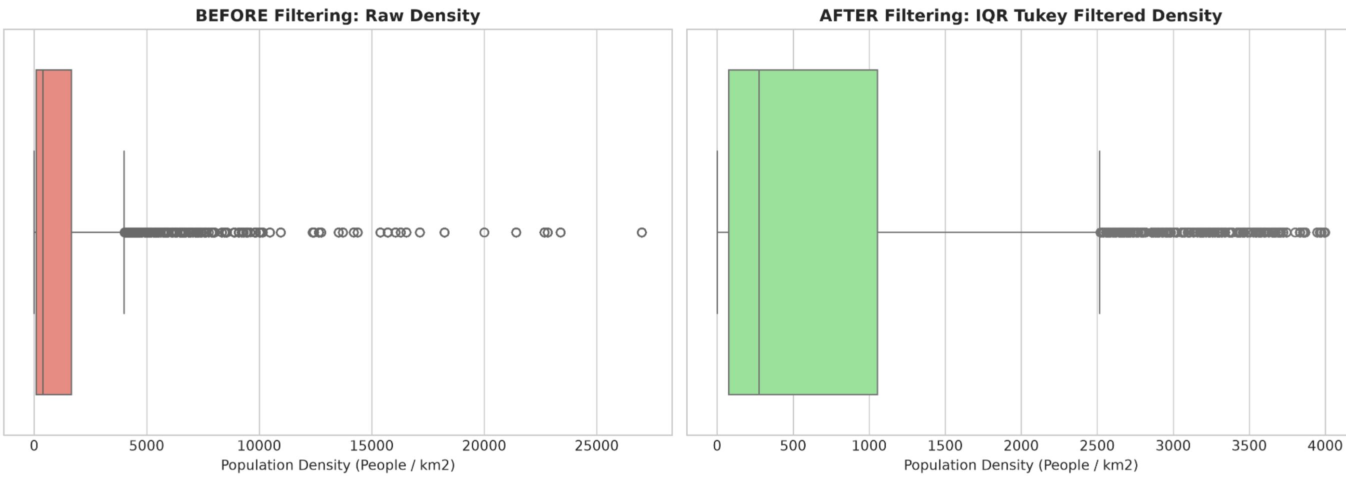

Figure 3. Before/after visual comparison of Density using the Tukey IQR filter.

The production script and the notebook's audit are consistent on the key result: the Tukey filter removes 77,566 observations, or 11.44% of the original dataset, leaving 600,447 records. The resulting CSV is the required cleaned-data artifact for Part 5.1.

Code script:

```
# Cell 8: Mathematical Outlier Filtering & Visual Before/After Comparison


target_feature = "Density"


# Tukey IQR Method

Q1 = df[target_feature].quantile(0.25)

Q3 = df[target_feature].quantile(0.75)

IQR = Q3 - Q1


iqr_lower = Q1 - 1.5 * IQR

iqr_upper = Q3 + 1.5 * IQR


df_iqr_cleaned = df[

    (df[target_feature] >= iqr_lower) &
```

```
    (df[target_feature] <= iqr_upper)
]

# Z-Score Method
z_scores = (
    (df[target_feature] - df[target_feature].mean()) /
    df[target_feature].std()
)

df_z_cleaned = df[abs(z_scores) <= 3.0]

# Outlier Audit Report
print(f"=== OUTLIER AUDIT REPORT FOR '{target_feature}' ===")
print(f"Original Record Count: {len(df):,} records")
print(
    f"Tukey IQR Range [{iqr_lower:.2f}, {iqr_upper:.2f}]: "
    f"Excludes {len(df) - len(df_iqr_cleaned):,} records"
)
print(
    f"Z-Score Threshold (|Z| <= 3.0): "
    f"Excludes {len(df) - len(df_z_cleaned):,} records"
)
```

```
# Before vs. After Visualization
fig, axes = plt.subplots(1, 2, figsize=(14, 5))

sns.boxplot(
    x=df[target_feature],
    ax=axes[0],
    color='salmon'
)
axes[0].set_title(
    f'BEFORE Filtering: Raw {target_feature}',
    fontsize=12,
    fontweight='bold'
)
axes[0].set_xlabel('Population Density (People / km2)')

sns.boxplot(
    x=df_iqr_cleaned[target_feature],
    ax=axes[1],
    color='lightgreen'
)
axes[1].set_title(
    f'AFTER Filtering: IQR Tukey Filtered {target_feature}',
    fontsize=12,
```

```
    fontweight='bold'
)
axes[1].set_xlabel('Population Density (People / km2)')

plt.tight_layout()

fig_filter_path = os.path.join(
    "..",
    "reports",
    "figures",
    "outlier_filtering_comparison.png"
)

plt.savefig(fig_filter_path, dpi=300)
plt.show()

print(
    f"[+] Saved visual outlier comparison artifact to: "
    f"{fig_filter_path}"
)
```

5.2 Final PDF and Reproducibility Checklist

- Final report filename: Module1 Homework Report.pdf
- Production script: src/clean_outliers.py

- Cleaned data output: data/cleaned_business_data.csv
- Notebook: notebooks/01_eda_and_data_dictionary.ipynb
- Figures: correlation heatmap, feature distributions, and outlier-filter comparison.
- Before final Canvas submission, push all files to the public GitHub repository and verify that the README contains reproduction instructions and that the PDF is downloadable.
- Per the assignment, have a classmate independently run the project from the README instructions and fix any errors before submission.